

4] In the formula $MgSO_4 \cdot xH_2O$, What does the x before the H_2O stand for?

A. The number of moles of anhydrous

B. The number of moles of hydrogen

C. The number of moles of water

D. The number of moles of hydrate.

5] Using the blue flame in hydrated salt experiment, then the value of ($H_2O\%$) will be :

A. Decrease

B. Increase

C. No effect

6] Which of the following Pressure will give you the highest boiling point of Acetone ?

A. 1.5 atm

B. 0.52 atm

C. 1 atm

D. 1.85 atm



7] Which of the following will not dissolve in water (H_2O) and form a precipitate?

A. $AgNO_3$

B. $CaCO_3$

C. CH_3CH_2OH

D. $NaOH$

8] What is the empirical formula if you have 83.0% carbon and 17.0% hydrogen? (Mm H=1 / C=12 g/mol)

A. C_2H_5

B. CH_4

C. C_3H_8

D. C_4H_{10}

9] The name of a balance with (± 0.0001 g) accuracy is:-

A. Analytical

B. Top loading

C. Two handle

D. Triple beam



~ The solutions:

[1] decrease the temperature

[2] decrease

[3] mass of $(H_2O) = 37.5\%$ — (د) نينا

M.Wt " = 18 g/mol

M.Wt (Mm) anhydrous = 379 g/mol

mass of anhydrous soln = mass hydrates - mass of water

$$= 100\% - 37.5\%$$

$$= 62.5\% \text{ or } (ج)$$

$$X = \frac{\text{mol of } (H_2O)}{\text{mol of anhydrous}} = \frac{\frac{\text{mass of } H_2O}{\text{M.Wt}}}{\frac{\text{mass of anhydrous}}{\text{M.Wt}}} = \frac{\frac{37.5}{18}}{\frac{62.5}{379}}$$

$$= 12.6$$

[4] the number of moles of hydrate

[5] decrease

[6] 1.85 atm

[7] $CaCO_3$

[8] $C = 83\%$ } mass C = 83 g } — m
 $H = 17\%$ } mass H = 17 g }

$$\text{mole C} = \frac{83}{12} = 6.9$$

$$\text{mole H} = \frac{17}{1} = 17$$



« نواتج » - ٢٠٢٠

[1] Identify a method to DECREASE the solubility of solute:-

- A. increase the temperature
- B. decrease the temperature
- C. increase the pressure
- D. stir the solution

[2] The calculated ratio between O/Mg in your experiment will be, if some of the product is "lost" before the final mass is measured?

- A. Be the same
- B. Increase
- C. Decrease

[3] If the % mass of H_2O = 37.5%, and the molar mass (Mm) of anhydrous salt = 379 g/mol. The value of water of crystallization (x) is: (Mm H_2O = 18 g/mol)

- A. 10
- B. 16
- C. 13
- D. 12



10] forgetting to dry the outside of the flask when you weigh the gas, then the molar mass of volatile will be :

- A. No effect
- B. Increase
- C. Decrease

11] A density of a liquid equal 1.82 g/L was evaporated if the temperature was 353.5 K at 798 mmHg then its molar mass is: ($R = 0.0821$)

- A. 60 g/mol
- B. 50.3 g/mol
- C. 42.3 g/mol
- D. 85 g/mol

12] Which of the following statements is correct?

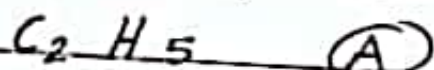
- A. Hold your face directly over the container when you smell a chemical substance.
- B. Return used chemicals to the stock bottles.
- C. When dealing with alcohols, heat with indirect flame by using water bath.
- D. It's allowed to withdraw samples from the stock solution bottles with your own pipet.



$$\begin{array}{r} \underline{C} \quad \underline{H} \\ 6.9 \quad 17 \\ \hline 6.9 \quad 6.9 \end{array}$$

C : H

$$(1 \quad 2.5) \times 2$$



[9] Analytical

[10] Increase

$$[11] \rho = 1.82 \text{ g/L}, T = 353.5 \text{ K}, R = 0.0821$$

 $P = 798 \rightarrow$ (atm) فحولها الى

$$[1 \text{ atm} = 760 \text{ mmHg}]$$

$$M.Wt = \frac{1.82 \text{ g} \times 353.5 \text{ K} \times 0.0821 \text{ L} \cdot \text{atm} \times 760 \text{ mmHg}}{\text{L} \times \text{mol} \times \text{K} \times 798 \text{ mmHg} \times 1 \text{ atm}}$$

[12] C

$$[13] d = \frac{\Delta m}{V} \rightarrow \frac{67.5 - 52.3}{25} = 0.6 \quad (A)$$

